Blade Concept Matrix

implicit-geometry-kernel
Companion to `docs/blade/concepts/taxonomy.md`

Abstract

A catalogue key names a blade concept. The Markdown page is canonical; figures below are example 09 cutaways unless noted.

1 Concept Catalogue

```
[W][X]-[Y]-[Z]-[R]      e.g.  2D-F-LC-0R    2D-F-LE-10R    2D-F-PA-10R
 |  |   |   |   +-- rib count (required NR): n = 0..10 (figures: 0R · 4R · 10R)
 |  |   |   +------ web anchoring (required): LC | LE | PA
 |  |   +---------- web orientation: F flapwise | CN chord-normal
 |  +-------------- spar-cap family: A B C D
 +----------------- web count: 0..5
```

Zero-web cores omit $[Y]$: `0A-LC-0R`, `0A-LE-10R`. Bare `2D-F` and omitted `0R` are invalid. `BladeConcept.from_key("2D-F-LC-0R")` / `resolve_system_type(...)` yields a frozen `SystemLayoutSpec`.

Table 1: Concept letters.

Part	Meaning	Examples
W	Shear-web count	`0 1 2 ... 5`
X	Spar-cap family	`A B C D`
Y	Web orientation	`CN F` (omit when $W = 0$)
Z	Web anchoring	`LC LE PA`
R	Rib count (required `NR`)	integer $0 \ldots 10$; figures `0R 4R 10R`

2 W — web count

Fixed $X = \mathrm{D}$, $Y = \mathrm{F}$, $Z = \mathrm{PA}$, $R = 10$. Two-web `2D-F-PA-10R` is under Z.

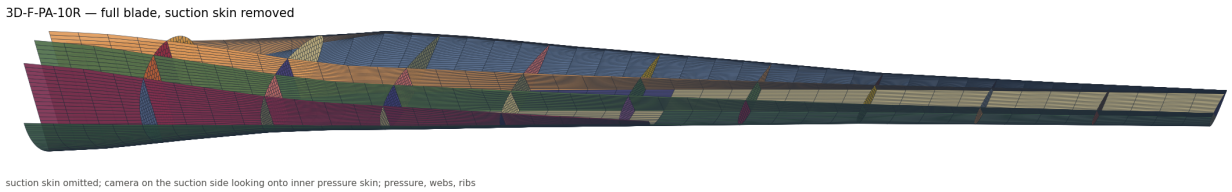


Figure 1: `3D-F-PA-10R`: three webs.

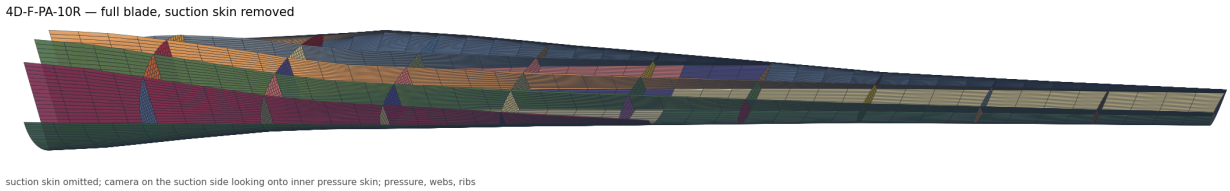


Figure 2: `4D-F-PA-10R`: four webs.

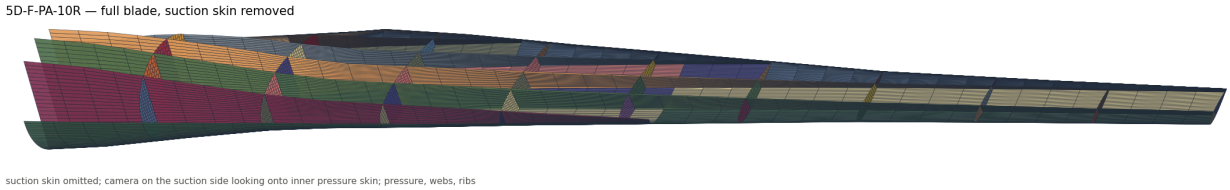


Figure 3: `5D-F-PA-10R`: five webs.

3 X — spar-cap families

Table 2: Spar-cap families.

X	Description
A	No spar caps: skin, webs, cores (and optional inserts) only.
B	**Single** fixed upper/lower cap pair on **one** chordwise band. Anchor: **pitching** (default $\sim \frac{1}{3}$ chord from LE) or **max_thickness**.
C	**Multiple** discrete upper/lower cap patches — one band per web — merged as `spar_cap_suction_surface` / `spar_cap_pressure_surface`.
D	**Continuous box** spar spanning leftmost to rightmost web (needs $W \geq 2$; no `1D-*`).

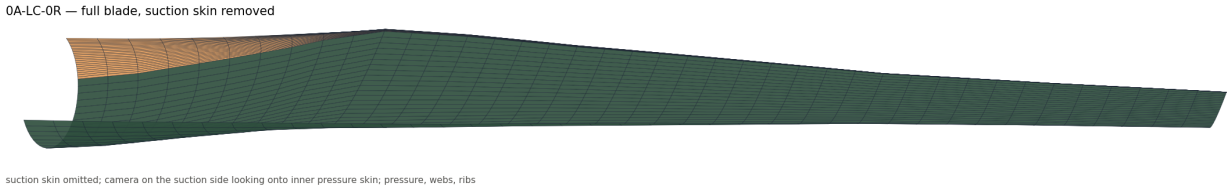


Figure 4: `0A-LC-0R`: family A, zero webs.

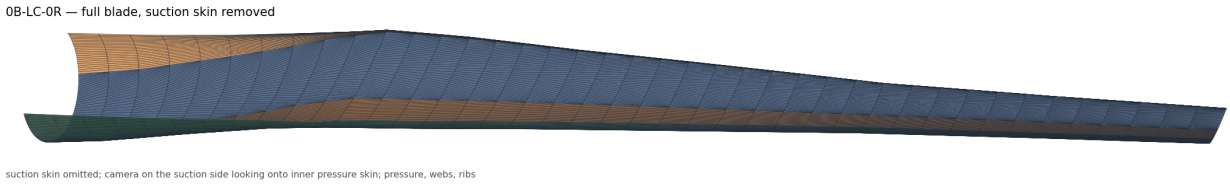


Figure 5: `0B-LC-0R`: family B, zero webs, one cap band.

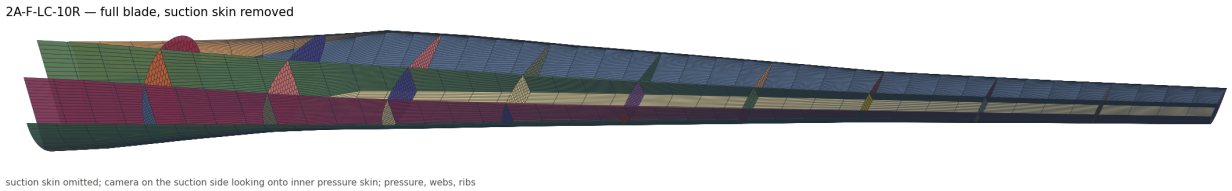


Figure 6: `2A-F-LC-10R`: family A, two flapwise webs, no caps.

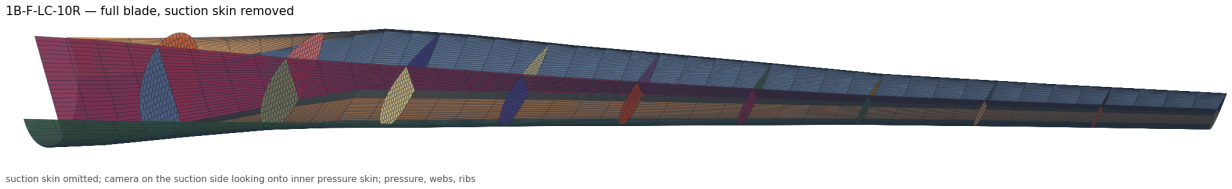


Figure 7: `1B-F-LC-10R`: family B, one flapwise web, one cap band.

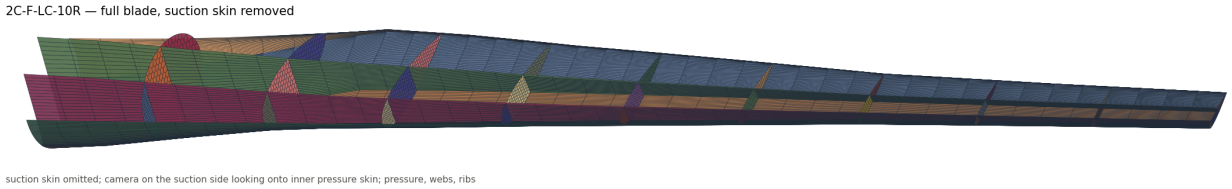


Figure 8: `2C-F-LC-10R`: family C, per-web cap bands.

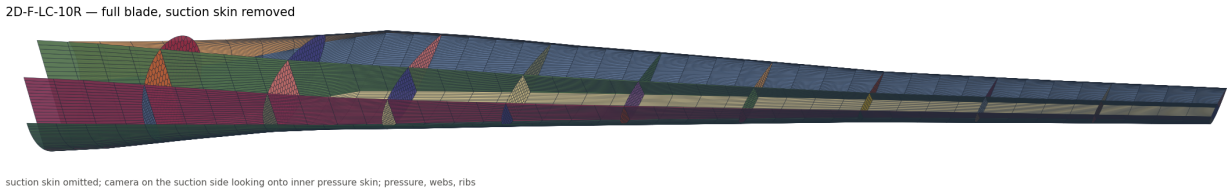
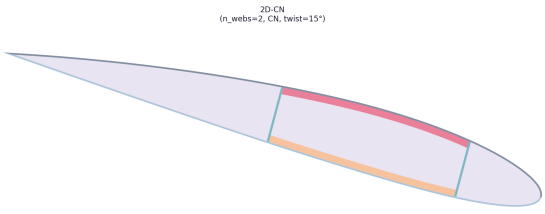


Figure 9: `2D-F-LC-10R`: family D, continuous box (also $Y = \mathrm{F}$, $Z = \mathrm{LC}$, $R = 10$).

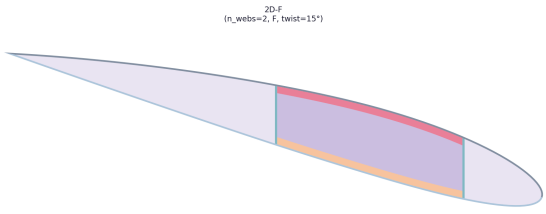
4 Y — web orientation

Table 3: Web orientation. Station slices; tilt is not obvious in a full-blade cutaway.

Y	Meaning
CN	Chord-normal in the **untwisted** section.
F	Vertical in the **twisted** blade frame (flapwise).



(a) `2D-CN`: chord-normal.



(b) `2D-F`: flapwise.

Figure 10: Web orientation at the station (`examples/demos/figures/structural_family_grids/`).

5 Z — web anchoring

Table 4: Web anchoring.

Z	Native unit	Span behaviour
LC	Chord fraction from LE	Stations slide with local chord $c(z)$.
LE	Metres from leading edge	Offset is frozen from the root chord; the web is omitted when it misses the inner skin.
PA	Blade-frame offset from the pitch axis	Constant x; primary slots are uniform interiors of the root LE–TE band. Extra slots continue that lattice (first extra at root TE).

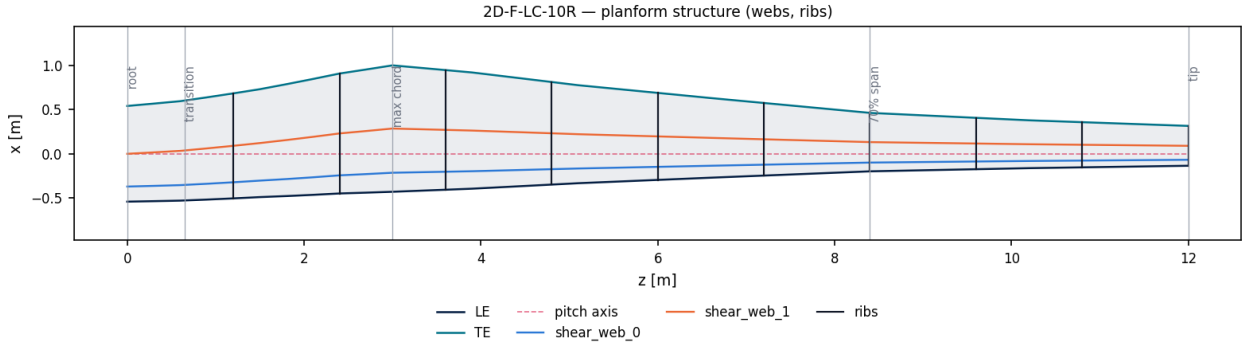


Figure 11: `2D-F-LC-10R`: loci follow local chord.

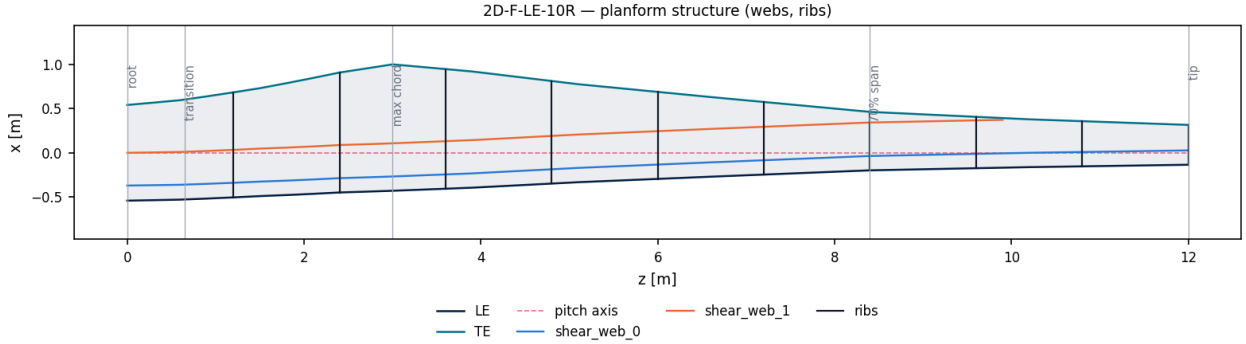


Figure 12: `2D-F-LE-10R`: leading-edge freeze.

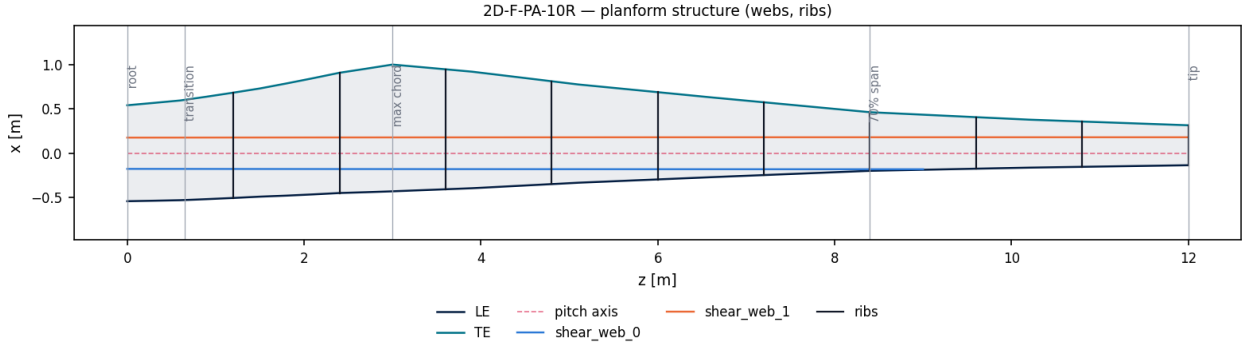


Figure 13: `2D-F-PA-10R`: pitch-axis freeze.

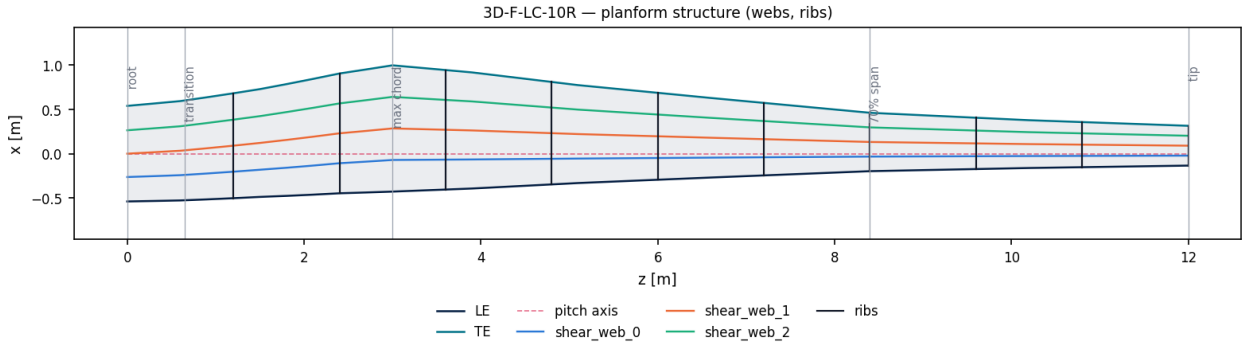


Figure 14: `3D-F-LC-10R`: three webs, local chord.

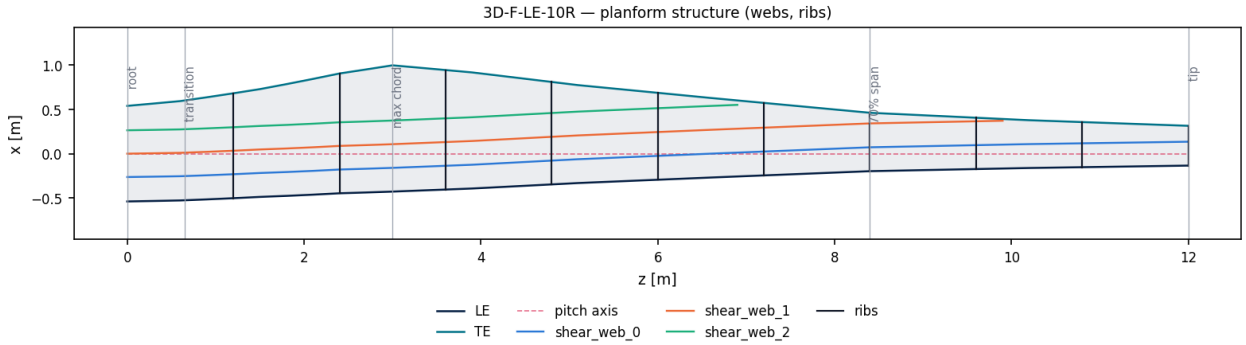


Figure 15: `3D-F-LE-10R`: three webs, leading-edge freeze.

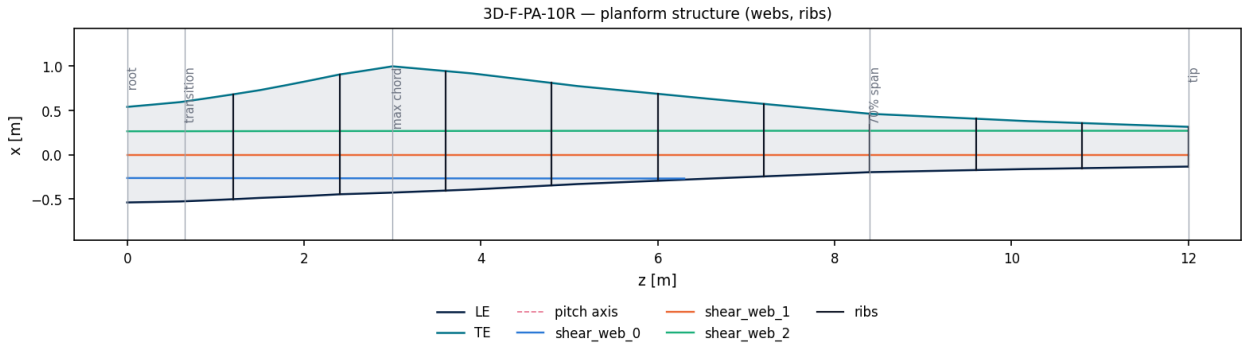


Figure 16: `3D-F-PA-10R`: three webs, pitch-axis freeze.

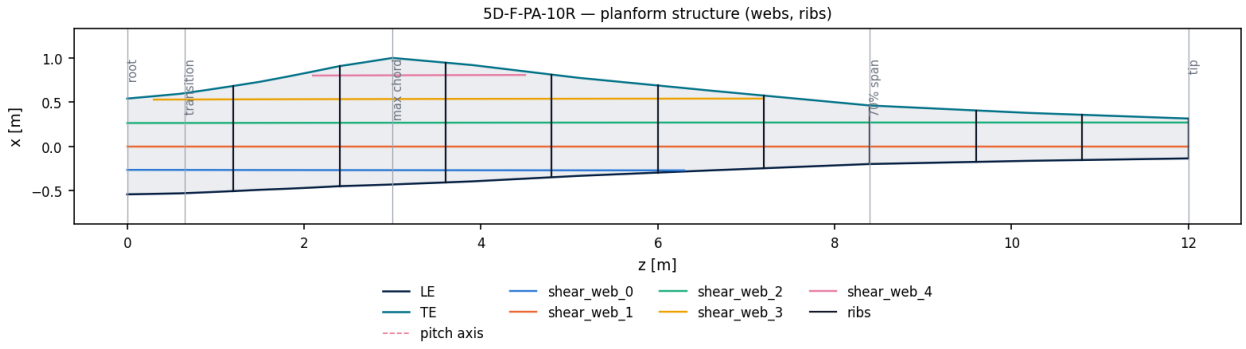


Figure 17: `5D-F-PA-10R`: five-web pitch-axis lattice.

6 R — rib count

Table 5: Rib count. $n = 0 \ldots 10$. This lookup shows none and four; `10R` is `2D-F-LC-10R` under X. `SYSTEM_TYPE_KEYS` enumerates `0R` only.

R	Meaning
n	Integer transverse-face count $0 \ldots 10$. Illustrated: `0R` none, `4R` four, `10R` ten.

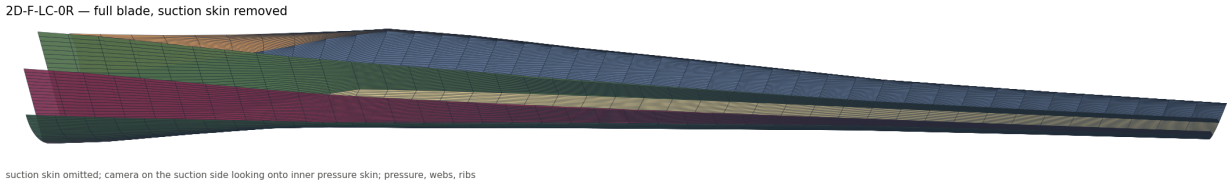


Figure 18: `2D-F-LC-0R`: no ribs.

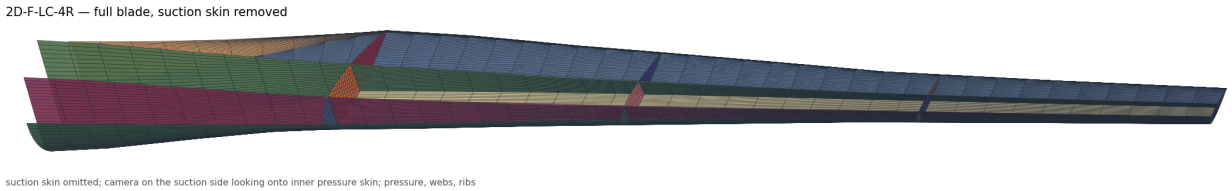


Figure 19: `2D-F-LC-4R`: four rib faces.

7 Stationwise, spanwise, transverse

A concept has three axes. `resolve_system_type` returns that whole spec.

Table 6: Blade-concept axes.

Axis	Letters	Fixes
Stationwise	WXY	Web count, cap family, tilt at a cut.
Spanwise	Z	Web $x(z)$: `LC` slides, `LE`/`PA` freeze.
Transverse	R	Rib-face count $0 \ldots 10$.

Table 7: Stationwise: 40 cores. Empty cells forbidden (no `0C`/`0D`, no `1D`). Attach `-Z-NR` to make a key.

W	`A-CN`	`A-F`	`B-CN`	`B-F`	`C-CN`	`C-F`	`D-CN`	`D-F`
0	`0A` (no Y)		`0B` (no Y)		—		—	
1	`1A-CN`	`1A-F`	`1B-CN`	`1B-F`	`1C-CN`	`1C-F`	—	—
2	`2A-CN`	`2A-F`	`2B-CN`	`2B-F`	`2C-CN`	`2C-F`	`2D-CN`	`2D-F`
3	`3A-CN`	`3A-F`	`3B-CN`	`3B-F`	`3C-CN`	`3C-F`	`3D-CN`	`3D-F`
4	`4A-CN`	`4A-F`	`4B-CN`	`4B-F`	`4C-CN`	`4C-F`	`4D-CN`	`4D-F`
5	`5A-CN`	`5A-F`	`5B-CN`	`5B-F`	`5C-CN`	`5C-F`	`5D-CN`	`5D-F`

Table 8: Spanwise: parent concept stays on `SectionCut.layout`. Webs that miss are **dropped**, not slid; live W/X is the station slice, e.g. `4D-F-PA-10R | W=1 X=C dropped=(0,1,2)`. Y, Z and R are not rungs. `0A`/`0B` are authored, not drop targets.

X	Floor (live webs)
A	nA $\to \cdots \to$ 1A
B	nB $\to \cdots \to$ 1B
C	nC $\to \cdots \to$ 1C
D	nD $\to \cdots \to$ 2D $\to$ 1C

Transverse: R is copied from the parent. `SYSTEM_TYPE_KEYS` is the 40 stationwise cores $\times$ $\{\text{LC}, \text{LE}, \text{PA}\}$ $\times$ `0R` (120 goldens). `1R`–`10R` resolve.